Advanced Computer Aided Design and Analysis

Module 1 - ABE 721: Agricultural Systems Modelling and Simulation

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Lecture Outline

1. 1. Geometric Modelling Techniques
2. 2. Systems Analysis Basics
3. 3. Farm Machinery Scheduling Problem
4. 4. Design of Farm Machinery Using CAD

What is Geometric Modelling?

Definition

The mathematical representation of the shape, structure, and spatial properties of physical objects within a computational environment.

Core Principles in Agricultural Engineering:

- Forms the absolute foundation of **Computer-Aided Design (CAD)**.
- Transitions conceptual schematics into precise digital prototypes.
- Exists in 2D or 3D Cartesian space using vector mathematics.
- Contains critical topological data required for downstream simulation (FEA, Kinematics, CAM).

Types of Geometric Models

1. Wireframe Models

- Vertices connected by edges.
- *Pros:* Fast processing.
- *Cons:* Ambiguous; no mass.

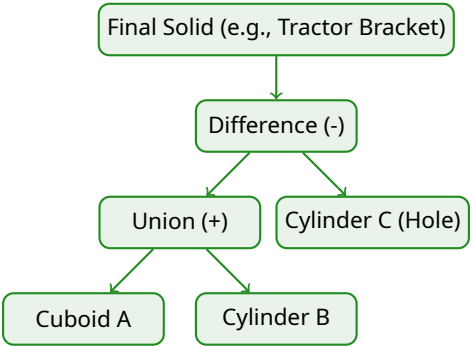
2. Surface Models

- Represents the ”skin” or boundary.
- Uses Bézier curves / NURBS.
- *Use:* Aerodynamics (plough moldboards, tractor canopies).

3. Solid Models

- Defines internal volume, mass, and outer boundary.
- Guarantees a physically realizable object.
- **Constructive Solid Geometry (CSG):** Boolean operations (+, -, $\cap$).
- **Boundary Representation (B-Rep):** Topological boundaries.

Constructive Solid Geometry (CSG)



Systems Analysis Basics

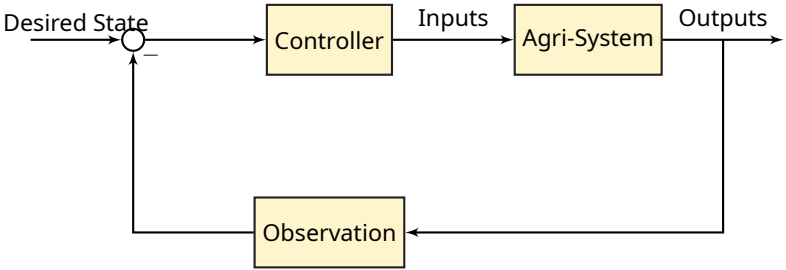
What is a System?

A set of interconnected, interdependent components working together within a defined boundary toward a specific goal.

System Architecture Components:

- **Inputs:** Resources entering from the environment (e.g., solar radiation, fuel, fertilizer).
- **Processes:** Physical/biological mechanisms transforming inputs.
- **Outputs:** Resulting consequences (e.g., crop yield, exhaust).
- **Feedback:** Output information routed back to influence inputs, crucial for control.

Closed-Loop Agricultural System



Taxonomy of Systems

Agricultural engineering deals with highly complex classes of systems:

- **Deterministic vs. Stochastic:**
  - *Deterministic:* Predictable outcomes.
  - *Stochastic:* Involves random variables (weather, disease). Agricultural systems are inherently stochastic.
- **Static vs. Dynamic:**
  - *Static:* Unchanging over time (equilibrium).
  - *Dynamic:* Evolving states requiring time-variant models ($\frac{dx}{dt}$).

Application: Farm Machinery Scheduling

The Problem

Allocating constrained resources (tractors, implements, operators) to minimize delays, reduce costs, and prevent biological yield penalties.

Modelling Approach (Operations Research):

1. **Define Objectives:** E.g., Minimize makespan or fuel cost.
2. **Identify Constraints:**
   - *Precedence:* Ploughing must occur before planting.
   - *Resources:* Tractor availability.
   - *Biology:* Planting windows and soil trafficability.
3. **Build & Run Scenarios:** Formulate as Mixed-Integer Linear Programming (MILP) or solve via heuristic simulation.

Design of Farm Machinery Using CAD

Modern engineering abandons trial-and-error drafting in favor of digital-first design and simulation.

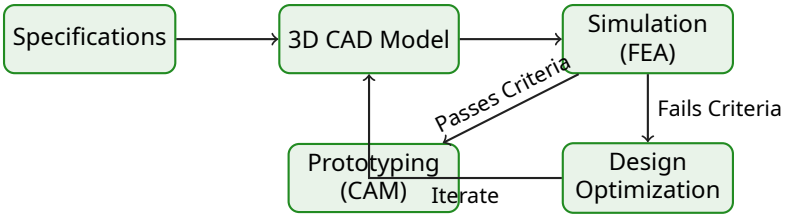
Computer-Aided Engineering (CAE) & FEA:

- **Finite Element Analysis (FEA):** Agricultural machinery faces extreme dynamic loads (e.g., hitting rocks).
- CAD models are ”meshed” to calculate structural stress using the stiffness matrix: $[K]\{U\} = \{F\}$.

Advantages:

- High precision and geometric tolerancing.
- Rapid, parametric modification.
- Cost reduction by avoiding physical destruction of prototypes.

Iterative CAD to CAM Workflow



Summary of Module 1

- **Geometric Modelling** (Wireframe, Surface, Solid) is the mathematical bedrock of digital agricultural engineering.
- **Systems Analysis** requires defining strict boundaries and mapping inputs, outputs, and feedback loops.
- **Scheduling** is an operations research problem designed to optimize constrained farming resources.
- **CAD and FEA** workflows radically accelerate machinery design and prevent structural failures prior to physical CAM prototyping.

End of Module 1

Questions & Discussion